

BANARAS HINDU UNIVERSITY

Under Graduate Semester III Examination, 2025-26 (NEP)

PHYSICS (SEC)**Paper Code: PHYSE31 : Computational Physics : Programming and Data Analysis**

Ref. No.: CECNF/N/2025-26/007

Time: $2\frac{1}{2}$ Hours

Maximum Marks: 70

Instructions: Answer five questions including Question 1 which is compulsory. Use Python programming language to address all the questions.

Question 1. Answer *any four* of the following questions: [4 × 5 = 20]

- (a) Program to print the multiplication table of 4 using ‘for’ loop.
- (b) Program to count digits in your Exam roll number.
- (c) Program to reverse the statement “We Are Happy”.
- (d) Write the value that gets printed, or write “error” if there is an error during the execution:

```
numbers = [4, 2, -1]
x = 1
for num in numbers:
    x = x * numbers
print(x)
```

- (e) Write the value that gets printed, or write “error” if there is an error during the execution:

```
x = [1, 2, 3]
x[len(x) - 1] = x[len(x) - x[0]] - x[1]
print(x[2])
```

Question 2.

- (a) Write a program to print: [6]
 - (i) your semester and course name.
 - (ii) your enrolment number.
 - (iii) name of your examination.

- (b) Find the data type of: [6 $\frac{1}{2}$]
 - (i) your course name.
 - (ii) your enrolment number.
 - (iii) name of your examination.

Question 3.

- (a) Write a python code to find the biggest of three numbers 'x', 'y' and 'z'. Thereafter, assign the values of your choice to the variables and obtain the results. [8]
- (b) Explain 'continue' and 'break' statements with an example. [4½]

Question 4.

- (a) Given `fruits = ["apple", "banana", "cherry", "date"]`: [6½]
- (i) Change the first element of the above list to orange.
 - (ii) Change the above list to tuple.
 - (iii) What is the difference between a list and a tuple?
 - (iv) Print each fruit in the list 'fruits'.
- (b) Discuss the following list functions with the help of examples: [6]
- (i) `len()`
 - (ii) `sum()`
 - (iii) `any()`

Question 5. Make a 4×4 array (of integral numbers) using `numpy`. [12½]

- (a) Slice a 2×2 submatrix out of it.
- (b) Find the mean of all the rows.
- (c) Find the standard deviation of all the columns.
- (d) Write these outputs to a file named 'int_mat.txt'.

Question 6. Given:

[12½]

```
Name      : ['Sam', 'Andrea', 'Alex', 'Robin', 'Kia']
Age        : [14, 25, 55, 8, 21]
Gender     : ['M', 'F', 'M', 'M', 'F']
Weight     : [45, 88, 56, 15, 71]
```

Use the above data to create a data frame and calculate the following:

- (a) Sort the dataframe by age in descending order.
- (b) Divide the dataframe based on gender.
- (c) What are three key differences between numpy and pandas?

Question 7. Given $x = (1, 2, 3, 4, 5, 6, 7)$, write a python program to draw the curves between x vs x^2 and x vs x^3 in a single figure. Add a title, axes label and legends on the plot. How do you change the legend fontsize in matplotlib? [12½]
